Daniele Iannotti

Curriculum Vitae

d.iannotti@ssmeridionale.it

Personal data

Name and Surname **Daniele Iannotti**

Place and year of birth **Italy, 1998**

Education

2023 – Present **PhD**, *Theoretical Physics*, SSM (*Scuola Superiore Meridionale*), Napoli, Italy

2020 – 2022 **Master Degree**, *Theoretical Physics*, Università degli Studi di Trieste, Italy

2017 – 2020 **Bachelor Degree**, *Physics*, Università degli Studi di Roma "La Sapienza", Italy

2012 – 2017 **Diploma**, Liceo Statale Scientifico Ettore Majorana, Latina, Italy

Experiences

2025 - Present **Founding Member**, *qmeets*, Online talks on recent breakthroughs in quantum physics for a broad audience

2024 - Present **Scientific Reviewer** for PRA, PRB, PRX Quantum, The European Physical Journal Plus, Annals of Physics

2024 - Present **Quantum Student Contributor** for Physics World Journal

2019 - Present **Academic Tutor**, Scuola Superiore Meridionale, Università degli Studi di Trieste and Università degli Studi di Roma "La Sapienza"

2025 - 2026 **Group meetings coordinator**, *for the QUANTUM SPACETIME and QUANTUM INFORMATION*, Università degli studi Federico II

2023 **Postgraduate student**, Neuroscience, SISSA, Trieste, Italy

2022 - 2023 **High school teacher**, Mathematics, Physics, IT

Conferences, workshops, schools and talks

Dec 2026 **Organizing Commitee**, *Zappatore – Recent Development in Theoretical Physics and its Neighbourhoods 8th edition*, University of Naples "Federico II" and NQSTI, Italy
21-22/12/2026

Oct 2026 **Invited talk for the Quantum Information and Gravity**, *Virginia Tech's group seminar*, Online, (*Talk*: Fermionic non-local magic)
19/10/2026

Oct 2026 **Workshop on Many-Body Quantum Magic**, Fondazione Bruno Kessler, Italy, (*Talk*: Non-stabilizerness and symmetries in chaotic many-body quantum systems)
05/10/2026-16/10/2026

Sept 2026 **Quantum Roundabout**, University of Nottingham, UK, (*Talk*: Nonlocal magic resources of fermionic Gaussian states)
09/09/2026-11/09/2026

Aug 2026 **2nd Workshop on random quantum circuits**, University of Cologne, Germany
17/08/2026-21/08/2026

Apr 2026 **ICTP-IBM School on Quantum Computing and Simulation**, ICTP,Trieste,Italy, (*Poster*: Non-stabilizerness and symmetries in chaotic many-body quantum systems)
20/04/2026-30/04/2026

Dec 2025 **Zappatore – Recent Development in Theoretical Physics and its Neighbourhoods 7th edition**, Dipartimento di Fisica "Ettore Pancini" Università di Napoli Federico II,Italy, (*Talk*: Non-stabilizerness and symmetries in chaotic many-body quantum systems)
19-22/12/2025

Nov 2025 **Quantum Highlights for Young Researchers**, NQSTI and CNR,Pozzuoli, Italy, (*Talk*: Haar measure and non-stabilizerness)
28/11/2025

Sept 2025 **IQIS 2025**, Università degli studi di Bologna, Italy, (*Poster*: Entanglement and Stabilizer entropies of random bipartite pure quantum states)
08-12/09/2025

Aug 2025 **Exact Solvability and Quantum Information**, Les Houches Summer School,France
04-29/08/2025

Jun 2025 **Organizing Commitee**, *Stupor Quanti II: The Wonders of Quantum Mechanics*, University of Naples "Federico II" and NQSTI, Italy
10/06/2025

Apr 2025 **QCTIP 2025 - Quantum Computing Theory in Practice**, Physics department of Freie Universität Berlin, Germany, (*Poster*: Entanglement and Stabilizer entropies of random bipartite pure quantum states)
23-25/04/2025

Feb 2025 **SFT 2025 - Lectures on Statistical Field Theories**, Galileo Galilei Institute,Florence, Italy
03-14/02/2025

Dec 2024 **Zappatore – Recent Development in Theoretical Physics and its Neighbourhoods 6th edition**, Dipartimento di Fisica "Ettore Pancini" Università di Napoli Federico II,Italy, (*Talk*: Interplay between entanglement and non-stabilizerness)
16/12/2024

Nov 2024 **First workshop on Many-body quantum magic (MBQM2024)**, Technology Innovation Institute (TII), Abu Dhabi, United Arab Emirates, (*Poster*: Non-local magic beyond many-body systems)
18/11/2024 - 20/11/2024

Sept 2024 **Conference TFI 2024: Theories of the Fundamental Interactions**, Università di Napoli Federico II
23/09/2024 - 25/09/2024

Sept 2024 **Conference SuperFluctuations 2024**, Università di Salerno, Italy, (*Poster*: Quantum resource variational principle)
16/09/2024 - 18/09/2024

Mar 2024 **Conference PAFT24 - Quantum Gravity and Information**, Lloyd's Baia Hotel Vietri sul Mare, Salerno,Italy
24/03/2024 - 27/03/2024

Feb 2024 **VI International Workshop on Information Geometry, Quantum Mechanics and Applications**, Universidad Carlos III de Madrid in Leganés, Spain, (*Talk*: Geometry of Quantum Resources)
20/02/2024 - 22/02/2024

Feb 2024 **TPI school on QFT and holography: entanglement and symmetries**, Theoretisch-Physikalisches Institut FSU Jena, Germany, (*Talk*: Geometry of quantum resources)
12/02/2024 - 16/02/2024

Visiting

April-March 2027 **Visiting Perimeter Institute**, *Canada*, Shiyu Zhou's research group, Tutor for "Quantum Matter" course

April 2026 **Visiting University of Jena**, *Germany*, Martin Gärttner's group, (*Talk*: Why you should care about magic: randomness, symmetries, locality and beyond)
07/04/2026 - 10/04/2026

Feb-Sept 2026 **Visiting University of Cologne**, *Germany*, Xhek Trukeshi's research group (*Talk*: Magic and Randomness: Abelian symmetries in chaotic many-body quantum systems)

Sept 2024 **Visiting ICTP**, *Italy*, Collaboration with Marcello Dalmonte's group
09/04/2024 - 13/04/2024

Jun 2024 **Visiting Università degli studi di Modena e ReggioEmilia**, *Italy*, Collaboration with Paolo Zanardi's group
27/06/2024 - 29/06/2024

Preprints and Publications

[1] Stefano Cusumano, Lorenzo Campos Venuti, Simone Cepollaro, Immacolata De Simone, Gianluca Esposito, Daniele Iannotti, Barbara Jasser, Jovan Odavić, Michele Viscardi, and Alioscia Hamma. "Non-stabilizerness and violations of CHSH inequalities". In: *Quantum* 10 (2026), p. 2214. DOI: 10.22331/q-2026-09-21-2214. arXiv: 2504.03351 [quant-ph].

[2] Daniele Iannotti, Lorenzo Campos Venuti, and Alioscia Hamma. "Van Hove singularities in stabilizer entropy densities". In: *J. Phys. A* 59.7 (2026), p. 075301. DOI: 10.1088/1751-8121/ae3cd6. arXiv: 2510.22253 [quant-ph].

[3] Daniele Iannotti and Elisa Vallini. "Colored Weingarten Calculus for Block-Unitary Ensemble". In: (Sept. 2026). arXiv: 2609.30385 [quant-ph].

[4] Beatrice Magni, Daniele Iannotti, and Riccardo Travaglino. "Entanglement spectrum and magic in higher-dimensional free fermionic systems". In: (Sept. 2026). arXiv: 2609.36008 [quant-ph].

[5] Barbara Jasser, Daniele Iannotti, and Alioscia Hamma. "A journey through Flatland: What does the antiflatness of a spectrum teach us?" In: (May 2026). arXiv: 2605.21664 [quant-ph].

[6] Daniele Iannotti, Beatrice Magni, Riccardo Cioli, Alioscia Hamma, and Xhek Turkeshi. "Non-Local Magic Resources for Fermionic Gaussian States". In: (Apr. 2026). arXiv: 2604.27049 [quant-ph].

[7] Daniele Iannotti, Angelo Russotto, Barbara Jasser, Jovan Odavić, and Alioscia Hamma. "Non-stabilizerness and U(1) symmetry in chaotic many-body quantum systems". In: (Mar. 2026). arXiv: 2603.28870 [quant-ph].

[8] Daniele Iannotti, Gianluca Esposito, Lorenzo Campos Venuti, and Alioscia Hamma. "Entanglement and Stabilizer entropies of random bipartite pure quantum states". In: *Quantum* 9 (2025), p. 1797. DOI: 10.22331/q-2025-07-21-1797. arXiv: 2501.19261 [quant-ph].

[9] Daniele Iannotti and Alioscia Hamma. "Geometric methods in quantum information and entanglement variational principle". In: *Int. J. Geom. Meth. Mod. Phys.* 21.10 (2024), p. 2440010. DOI: 10.1142/S0219887824400103. arXiv: 2403.13102 [quant-ph].

[10] Daniele Iannotti and Antonio Pittelli. "Twisted Index on Hyperbolic Four-Manifolds". In: *Lett. Math. Phys.* 114.2 (2024), p. 39. DOI: 10.1007/s11005-024-01788-x. arXiv: 2307.11634 [hep-th].

Awards, Honours and Grants

2026 **Research Grant** for Young Researcher and Innovator by COST Action CA23130
2026 **Research Grant** in Germany by DAAD
2026 **Research Grant** for Short visiting period by COST Action CA22113
2025 **Honours** Young ISSNAF Mentoring Program
2020 – 2022 **Scholarship** awarded by ARDISS
2020 **Award** "Premio di Laurea" by LazioDisco
2017–2020 **Scholarship** awarded by LazioDisco

Outreach activities

2025-2026 Penne Amiche della Scienza: Italian project similar to *Letters to a Pre-Scientist*
2024-2025 European Researchers' Night in Naples: theatrical dialague about quantum information
Apr 2024 Parla Potabile: quantum games and outreach
Apr 2024 Ludoteca quantistica and Quantum treks: quantum games and outreach
Sept 2022 Trieste Next: stand about physics and litterature
2019-Present Outreach activities and lectures at Liceo Scientifico E. Majorana, Latina, Italy

Languages

Italian **Mother tongue**, English **C1**, French **B1**